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Risk-Based Inspection Review - ATU-100

Executive Summary

This Corporate Summary, authored by Dr. Helen Marchetti and Jennifer Walsh, documents the first RBI review for ATU-100. The review ensures compliance with QSCP-CIS-001 Rev 0, which mandates RBI-based intervals, CML re-baselining, and FFS by a qualified engineer. This assessment incorporates the 2016 full-unit survey LNDE-RPT-20160118-004-R0 and the damage mechanism review QSCP-DMR-2014-002, which addressed the DEA strength increase MOC.

Damage Mechanism and Inspection History

Per QSCP-DMR-2014-002, the increase in acid-gas loading is expected to raise corrosion rates on the rich side. Consequently, rich amine circuits have moved to a 3-year interval with additional CMLs placed downstream of the let-down valve. Damage mechanisms were evaluated against API RP 571. Previous thickness data from LNDE-RPT-20160118-004-R0 established baseline intervals for Class 1 and Class 2 systems, including the 10-year cycle for Class 2 piping and 3-year cycle for the regenerator overhead.

Piping Technical Data

The review specifically addressed reflux lines 3-RF-101 and 2-RF-102. Line 3-RF-101 is A312 TP316 with an OD of 88.9 mm, t_nom of 5.49 mm, and t-min of 2.41 mm, designed for 517 kPa(g) at 104 °C. Line 2-RF-102 is A312 TP316 with an OD of 60.3 mm, t_nom of 3.91 mm, and t-min of 2.16 mm, also designed for 517 kPa(g) at 104 °C. All future inspections must adhere to the data requirements and code maxima defined in QSCP-CIS-001 Rev 0.

Sign-off	Name	Function	Date	Signature
Prepared	Dr. Helen Marchetti	Corporate Asset Integrity Manager, P.Eng.	2018-11-19	_____
Prepared	Jennifer Walsh	Inspection Coordinator, API 570 inspector	2018-11-19	_____